Ayush Shinde

FORWARD DEPLOYED ENGINEER

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SUMMARY

Forward Deployed Engineer with 2.5+ years of experience across machine learning, AI engineering, and production systems. Experienced in building LLM agents, graph-aware RAG pipelines, and enterprise API integrations, with a focus on reliability, evaluation, and deployment. Previously worked on predictive maintenance models and large-scale IoT data pipelines, bringing hands-on experience across AI development, data engineering, and cloud infrastructure.

TECHNICAL SKILLS

Programming & APIs: Python, SQL, FastAPI, REST APIs, ServiceNow REST APIs

Generative AI & Agents: LLM Agents, Tool Calling, RAG, Graph-Aware RAG, LangChain, LangGraph, FAISS, Qdrant, Guardrails AI, Semantic Search, LLM Evaluation

Machine Learning: PyTorch, scikit-learn, XGBoost, LSTM, Anomaly Detection, Time-Series Modeling, Feature Engineering

Cloud & MLOps: AWS (S3, EMR, SageMaker), Azure AKS, Docker, Kubernetes, Git, GitHub Actions, CI/CD, MLflow

Data Engineering & Enterprise Systems: PostgreSQL, MySQL, PySpark, dbt, DuckDB, ServiceNow CMDB, RBAC, Human-in-the-Loop, Tableau

EXPERIENCE

ServiceNow | TX | Forward Deployed Engineer **Feb 2026 - Present**

- Implemented FastAPI services connecting LLM agents to ServiceNow APIs for P1/P2 incident intake, CMDB asset lookups, and ticket updates, reducing incident MTTR from 45 minutes to under 10 minutes.
- Designed a graph-aware RAG workflow using CMDB asset relationships to retrieve relevant infrastructure context during incident triage, achieving 92% classification accuracy.
- Added RBAC, custom validation checks, and human approval for destructive agent actions to prevent unauthorized production changes.
- Established a 230-scenario evaluation suite covering incident classification, CMDB dependency retrieval, and agent tool execution to test for incorrect classifications, missing asset relationships, and unsafe actions before deployment.

Siemens | India | ML Engineer **Aug 2022 - Jul 2024**

- Trained LSTM and XGBoost models to detect early signs of equipment failure across 5 manufacturing plants, achieving 94% fault detection accuracy and reducing unplanned downtime by 27%.
- Engineered PySpark pipelines processing 500GB+ of sensor data from 10,000+ IoT devices, reducing data preparation time by 60% for downstream ML workloads.
- Deployed containerized inference services using Docker and Azure AKS to monitor anomalies across 2,000+ industrial assets, reducing model response time by 40%.
- Migrated on-premise ML workloads to AWS S3, EMR, and SageMaker to simplify deployment and scale data processing.
- Created Tableau dashboards for 50+ plant managers to track equipment health, failure patterns, and maintenance plans.
- Automated ML build and deployment workflows with GitHub Actions, cutting deployment errors by 50% and release cycles by 40%.

PROJECTS

RAG Support Insights & LLM Evaluation | Python, LangChain, FAISS, Ollama, scikit-learn, Streamlit

- Built a RAG-based support triage system over 48K tickets using LangChain, FAISS, and Llama 3.2 (Ollama), with FastAPI and Streamlit for issue classification and source-grounded next-step recommendations.
- Introduced a two-stage abstention gate that improved accuracy on answered tickets from 81.5% to 91.4% at 87% coverage across 200 unseen queries, while escalating 98% of out-of-scope inputs.

TrialScope: Clinical-Trial Failure Prediction Platform | Python, XGBoost, dbt, DuckDB, LangGraph, MLflow

- Developed an end-to-end ML platform using 30K ClinicalTrials.gov trials to predict early termination and delays, with SHAP explanations, a LangGraph SQL agent, and an interactive Streamlit dashboard.
- Evaluated XGBoost models trained only on information available at trial start, achieving 0.71 AUROC versus a 0.64 baseline and identifying 2.7x more terminated trials than random selection within the top 10% of risk scores.

EDUCATION

Master of Science in Data Science | University of Texas at Arlington, Arlington, TX **May 2026**

B.E. in Electronics & Telecommunication Engineering (AI & ML Honors) | University of Mumbai **Aug 2024**

CERTIFICATIONS

- **AWS Certified Machine Learning Engineer - Associate**
- **Databricks Fundamentals**
- **Google Advanced Data Analytics**